

f00tils

*Freestanding assembly GNU userland * <https://f00.sh>*

STANDARD OPERATING PROCEDURE

Document: SOP-F00-OPS-001
Date: 2026-07-27
Version: v0.16.7
Standard: NASA NAI 1450.3 memorandum layout

Reply to Attn of: OPS/F00

TO: Maintainers and Release Operators
FROM: Project Engineering
SUBJECT: f00tils operator standard operating procedure (clone, build, gates, install, release)
REF: AGENTS.md, README.md, asm/Makefile, install.sh,
scripts/render-brand-assets.py, .github/workflows/release.yml, v0.16.7

Bottom line: On Linux x86-64, from a clean tree, run `cd asm && make && make check` before you merge. Run `make hot` / `make speed` before you claim a speed win. Ship with `tarball` + `install.sh` as the primary path. Keep `--core` byte-identical to GNU and faster on wall time and CPU. Every release regenerates color-PTY screenshots (`scripts/render-brand-assets.py`) and ships NASA SOP + release memo on the GitHub Release.

1. Purpose

This memorandum is the living operator handbook for f00tils (binary f00 / tools f00-*). It states how to clone the repository, build the freestanding multicall, run quality gates, install, and ship a release.

2. Scope

This SOP covers full project operations. It does not replace man pages, README narrative, or the scoreboard on <https://f00.sh>. Those remain the public ship surfaces.

- repository layout and product laws
- host prerequisites
- x86-64 build and boring-solid gates
- hot-path and speed law
- aarch64 freestanding port
- local and end-user install

- release, packaging, and site install sync
- user-facing documentation rules

3. Product identity

Item	Value
Product name	f00tils
Binary / multcall	f00
Tool names	f00-* (bare names when replace is on)
Site	https://f00.sh
Repository	github.com/theesfeld/f00
License	MIT only
Primary target	Linux x86-64 freestanding (NASM, no libc)
Port path	Linux aarch64 freestanding (GNU as, asm/port/aarch64)
Surface	115 tools (coreutils + grep + findutils + diffutils)

4. Product laws (non-negotiable)

Language purity: no Rust, C application code, libc, or polyglot product dependencies in first-party product code under asm/. Shell is allowed only for bootstrap, install, packaging, and benches.

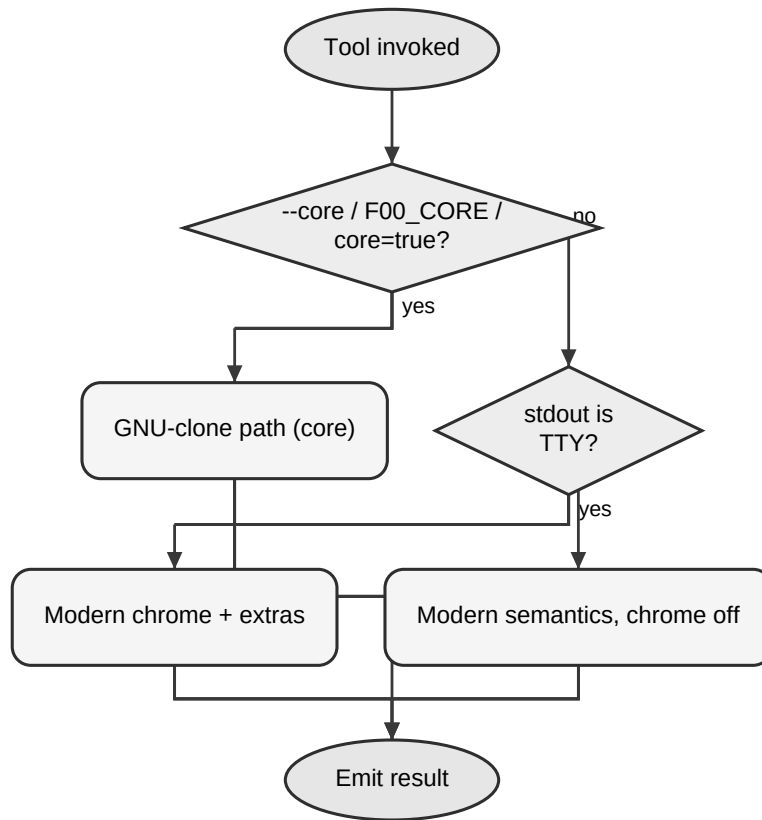
4.1 Clone first (--core). Every covered GNU tool has a f00-* counterpart. Under --core, match the GNU tool for scripts: flags, exit codes, and byte-identical stdout/stderr for the same inputs.

4.2 --core must win on resources. On the core path, f00 must beat the GNU tool on wall time and CPU (user+sys). Correct-but-slower is not done.

4.3 Modern is amazing (default). Non--core is never a pale GNU subset: themed chrome, better layout, icons where relevant, --json/--csv, and extra power.

4.4 One binary. Multicall by argv0 (f00-grep, grep, ...).

Figure. Script vs interactive path



5. Repository layout

Path	Role
asm/	Product source, Makefile, man pages, benches
asm/port/aarch64/	aarch64 freestanding multcall
site/	https://f00.sh (GitHub Pages) and site install.sh
install.sh	Root installer (must match site/ and scripts/)
packaging/	AUR, nfpdm (deb/rpm/arch)
Formula/	Homebrew formula
docs/	Compliance, UX, progress, this SOP
file_id.diz	Release scene card (GitHub Release asset only)
scripts/	Release, package, and bench generators

6. Prerequisites

6.1 x86-64 host (primary): Linux x86-64; nasm; ld (binutils); make; python3; GNU coreutils/grep/findutils/diffutils on PATH for parity and speed comparison.

6.2 aarch64 port (optional): aarch64-linux-gnu-as and aarch64-linux-gnu-ld; qemu-aarch64-static (or equivalent) for smoke on x86-64 hosts.

7. Procedure - clone and build (x86-64)

Do this on every new worktree before you edit product code. Root convenience targets: make (asm), make check, make aarch64, make install.

7.1 Clone or update the repository.

```
git clone git@github.com:theesfeld/f00.git
cd f00
git checkout main
git pull --ff-only
```

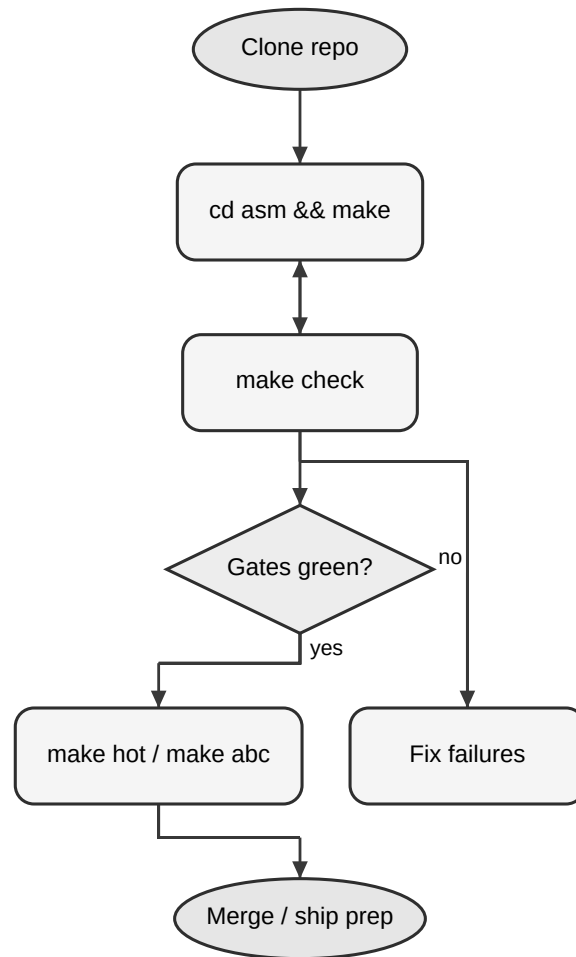
7.2 Enter the product tree and build the multicall.

```
cd asm
make
```

7.3 Confirm the binary runs.

```
./f00 --version
./f00-ls --version
```

Figure. Build and gate flow



8. Procedure - quality gates

8.1 Boring-solid (required before merge).

```
cd asm  
make check
```

8.2 Hot path (real-work wall + CPU). Use when you change sort, ls, or other hot tools.

```
cd asm  
make hot
```

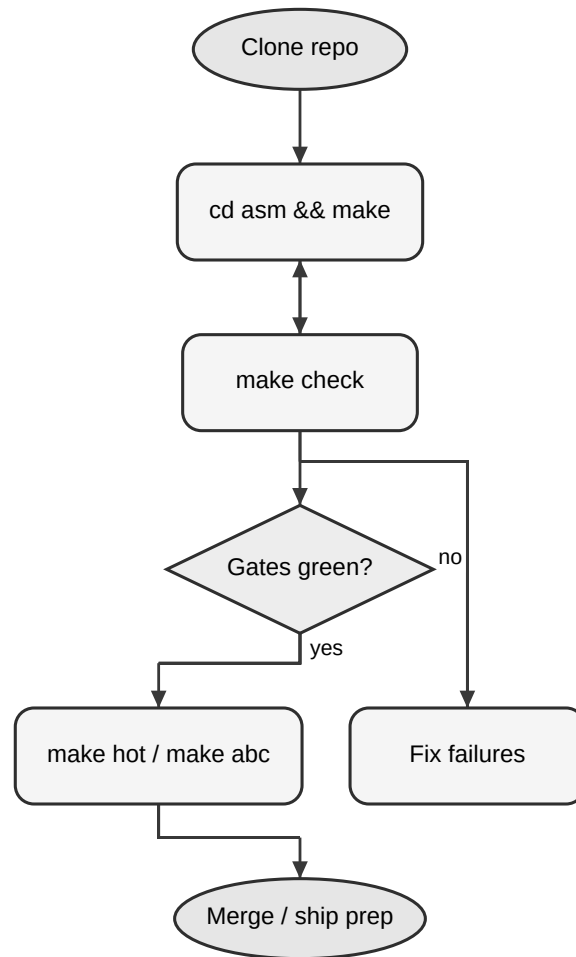
8.3 Speed law (wall + CPU). Optional each commit; required for law-2 claims. Includes hot.

```
cd asm  
make speed
```

8.4 Structural speed proof (stress).

```
cd asm  
make speed-stress
```

Figure. Build and gate flow



Target	Meaning
make / make all	Build f00 + tool links
make smoke	Minimal runtime smoke
make parity	GNU --core parity battery
make check	smoke + parity (default bar)
make hot	Real-work wall+CPU + stdout assert
make speed	wall+CPU law (includes hot)
make speed-stress	Repeated full-speed-gate proof
make aarch64	Build aarch64 multcall
make aarch64-smoke	qemu smoke of aarch64 binary

9. Procedure - aarch64 port

The aarch64 path is shipped and grows toward full surface. It is not the default CI/release install path today.

9.1 Install the cross assembler and linker, and qemu-user if you build on x86-64.

9.2 Build and smoke.

```
cd asm
make aarch64
make aarch64-smoke
```

9.3 On failure, fix the port under asm/port/aarch64. Do not weaken x86-64 gates to paper over port gaps.

10. Procedure - local and end-user install

Default installer places multcall f00, all f00-* names, and bare tool names in INSTALL_DIR so replace wins on PATH.

10.1 Install from a local build.

```
cd asm
make install
# or
F00_LOCAL=/path/to/f00/asm bash /path/to/f00/install.sh
```

10.2 End-user install (primary ship story).

```
curl -fsSL https://f00.sh/install.sh | bash
```

10.3 Knobs: F00_VERSION=vX.Y.Z (pin tag); F00_SUPERSEDE=0 (f00-* only);
INSTALL_DIR=~/.local/bin; F00_LOCAL=...; F00_MAN=1 (man pages, default on).

10.4 Config.

```
f00
f00-config
f00-config replace off
```

11. Procedure - keep installers identical

Root install.sh is the source of truth. Site and scripts copies must match byte-for-byte. Run after any edit to install.sh and before release.

11.1 Sync from repository root.

```
make sync-install
```

11A. Procedure - regenerate screenshots (every release)

Public screenshots must match the shipped binary and explain features (multi-command sessions), not decoration one-liners. Capture uses a real PTY and forced color environment.

11A.1 Build: `cd asm && make` (binary version string must match the release).

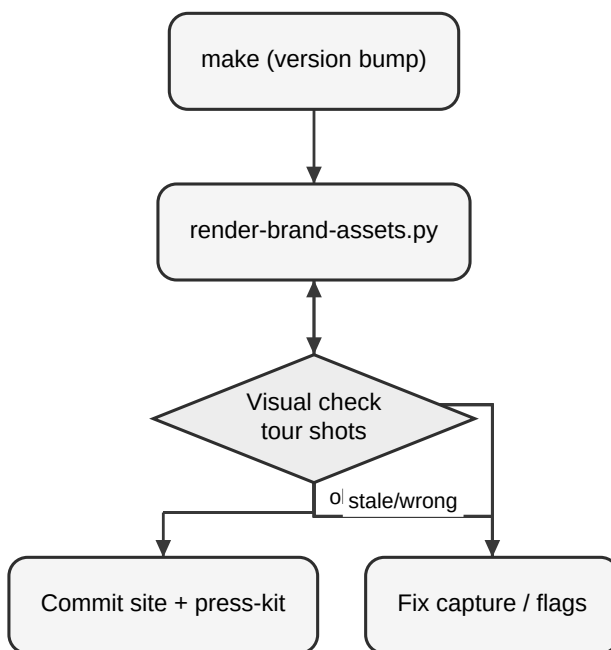
11A.2 Regenerate: `python3 scripts/render-brand-assets.py`

11A.3 Confirm outputs under `site/assets/screenshots/`, `press-kit/screenshots/`, `docs/images/`.

11A.4 Confirm site Feature tour (`#tour`) captions still match the shot set.

11A.5 Commit screenshots with the release; never leave stale version strings in `hero.png`.

Figure. Screenshot regen on ship



12. Procedure - release and ship

Primary release story: GitHub Release tarball + `install.sh`. `deb/rpm/AUR/brew` are secondary. Release assets must include `file_id.diz`, `sop-f00tils-ops.pdf`, and `memo-release-VERSION.pdf` (enforced by `.github/workflows/release.yml`).

12.1 Ensure main is green: build, make check, and speed gates as required for the change set.

12.2 Update version metadata so the binary, man pages, README pin examples, and CHANGELOG agree on `vMAJOR.MINOR.PATCH`.

12.3 Move Unreleased notes in `CHANGELOG.md` into a dated version section.

12.4 Refresh root `file_id.diz` for this version. Commit it. Attach it as a GitHub Release asset. Do not spotlight it on the website.

12.5 Sync installers (`make sync-install`).

12.6 Confirm man page(s), `README.md`, and site match behavior.

12.7 Create an annotated tag and push it.

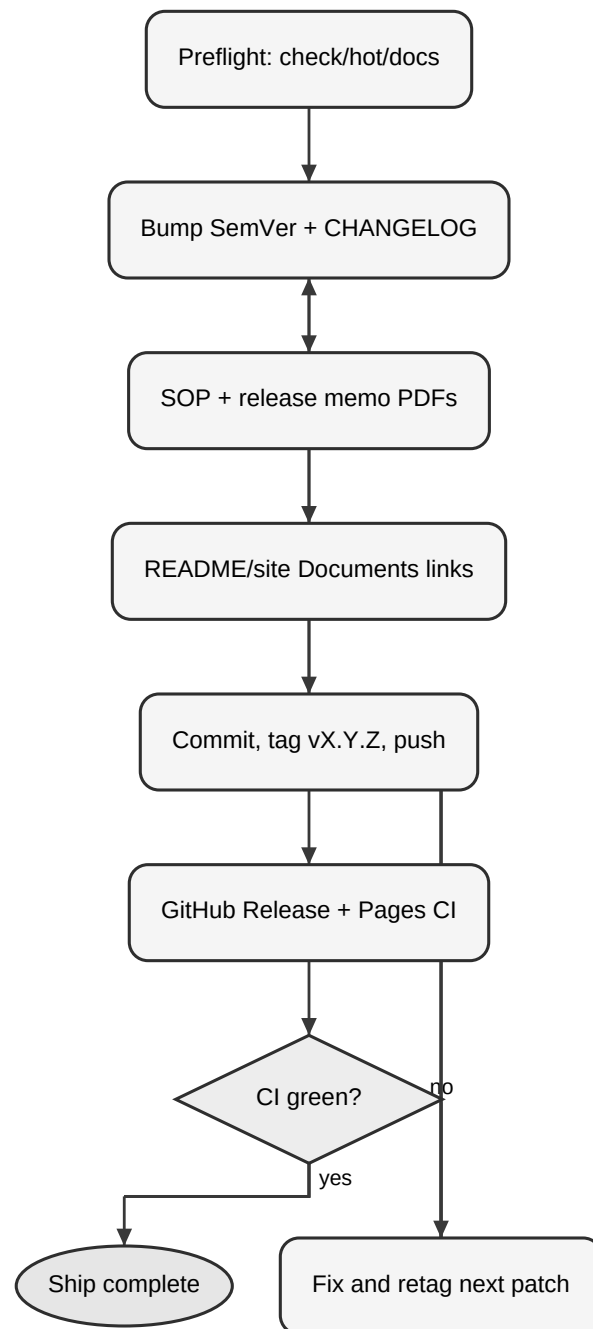
```
git tag -a vX.Y.Z -m "vX.Y.Z"  
git push origin vX.Y.Z
```

12.8 Release workflow (tag v*) builds Linux x86-64 ASM, runs smoke/parity/speed-gate, verifies binary version, packages tarball, publishes GitHub Release.

12.9 Smoke-test the public install path. Do not move or reuse a published tag for different bits.

```
curl -fsSL https://f00.sh/install.sh | F00_VERSION=vX.Y.Z bash  
f00 --version
```

Figure. Release and deploy flow



13. Procedure - secondary packaging

Use packaging scripts under scripts/ and configs under packaging/ and Formula/ when you intentionally ship secondary channels. They must not replace tarball + install.sh as the default install story.

- scripts/build-linux-packages.sh
- scripts/gen-aur-pkgbuild.sh / scripts/publish-aur.sh
- scripts/gen-homebrew-formula.sh / scripts/publish-homebrew-tap.sh
- scripts/sync-package-manifests.sh

14. Procedure - GitHub Actions constraints

- Use only Node 24 JS action majors (checkout@v6+, upload-artifact@v6+, download-artifact@v7+, softprops/action-gh-release@v3+, Pages @v5/@v6 as house rules specify).
- Do not reintroduce Node 20-era pins (checkout@v4, upload-artifact@v4, action-gh-release@v2). Deprecation annotations are a defect.

15. Procedure - user-facing text

- Name the project f00tils in narrative. Keep command names f00 / f00-.*.
- Procedures, man pages, and install how-to: ASD-STE100 (Simplified Technical English).
- Public story (README lead, website): NASA Stylebook + AP habits.
- Ship narrative surfaces: README + <https://f00.sh>. Docs under docs/ are optional depth.

16. Daily operator checklist

- Branch from main; keep PRs small and Conventional Commits-friendly.
- Edit only assembly under asm/ for product logic.
- cd asm && make && make check before you open or update a PR.
- Run make hot / make speed when the change touches core-path performance or hot tools.
- Update man pages and site/README when user-visible behavior changes.
- Never claim --core parity without parity evidence.
- Never claim a speed win without gate evidence.

17. Failure responses

Symptom	Action
make fails	Fix asm build. Confirm nasm/ld. Clean with make clean.
make check fails	Read smoke/parity logs. Restore --core GNU match.
make hot/speed fails	Profile the tool. Do not ship a slower core path.
install.sh mismatch	Run make sync-install. Re-diff site and scripts.
Release CI red	Match tag to binary version. Re-run gates locally.
aarch64 smoke fail	Fix port. Keep x86-64 gates independent.
Wrong action majors	Upgrade to Node 24-era pins per house rules.

18. References

- AGENTS.md - product laws and layout
- README.md - public install and modes
- CONTRIBUTING.md - contributor build bar
- asm/Makefile - build and gate targets
- install.sh - primary installer
- .github/workflows/release.yml - tag release pipeline
- https://f00.sh - site, scoreboard, install entry
- site/bench/suite.json - suite bench data
- ~/.grok/skills/memo-pdf - house memo PDF renderer used for this document

19. Approval and control

This SOP is controlled under docs/. Update it when gates, install paths, or release mechanics change. Prefer short procedural edits over narrative expansion. Regenerate the PDF with the memo-pdf skill (nasa-sop type); do not freestyle layout.

Project Engineering

f00tils / OPS

Document: SOP-F00-OPS-001

Date: 2026-07-27

Enclosure: None. Companion sources are in-repository paths listed in section 18.

cc: Maintainers, Release Operators